

The Last Mile Is an Idea: Cognitive Operators, Proposer Strength and the Limits of Office-Scale Verified Search

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Abstract

Verified search — a language model proposing programs, a hard evaluator scoring them, selection keeping the best — is how machine systems have recently moved mathematical records. The published systems vary the scaffolding (islands, ensembles, novelty machinery) and report the wins; none measures, with controls, which parts of the *proposer’s cognition* matter. We instrument a minimal FunSearch-style loop at office scale (a 30B-parameter local model on a laptop, 120–600 verified samples per run) with three operators isolated in our earlier work on open-ended generation [2]: a *schematic notebook* the model writes and carries instead of verbatim elite programs, a *named obstacle* (agenda), and *behavioural repulsion* from constructions already found. On six construction problems from the AlphaEvolve repository [1], a pre-registered 5×6 factorial with a replicate shows: the composed operators improve the fraction of the seed-to-record gap closed in 5 of 6 problems (+0.20 pooled, $p = 0.0625$ at the $n=6$ exactness floor); behavioural repulsion raises functional diversity at the minimum attainable p (+0.38, $p = 0.0156$) and collapses without the anchoring memory (three collapses alone, none composed [TODO: check]); and at larger budgets the composition scales while repulsion alone plateaus. Swapping the proposer for a frontier model (Claude Opus 5) under the same loop closes in tens of samples what the local model cannot close in hundreds, and exceeds the value AlphaEvolve reported on the beat-the-average problem (0.3897 vs 0.3890) with 300 samples; tripling the verifier’s time budget moves this by +0.0002. Across proposers, grids and time budgets the search stalls at $\approx 92\%$ of the best known value, whose record construction evidently lies outside the sparse-atom family every proposer writes: the remaining mile is an idea, not a budget. We release the harness, every candidate, and the pre-registrations.

1 Introduction

[TODO: Frame: verified search moved records (FunSearch, AlphaEvolve, ShinkaEvolve); what nobody measured (operator factorial; proposer curve under fixed loop); our question and contributions list.]

Contributions. (1) A pre-registered factorial of cognitive operators inside verified search, with a replicate. (2) The anchoring law, fifth independent appearance: repulsion without memory breaks — as prompt-only collapse here, echoing DPO and consolidation results in [TODO: cite paper 2]. (3) The scaling inversion: composition scales with budget, pure repulsion plateaus. (4) The first controlled proposer-strength curve under a fixed loop, local 30B vs frontier API. (5) An honest calibration of office-scale verified search: 85–99% of records at 120–600 samples;

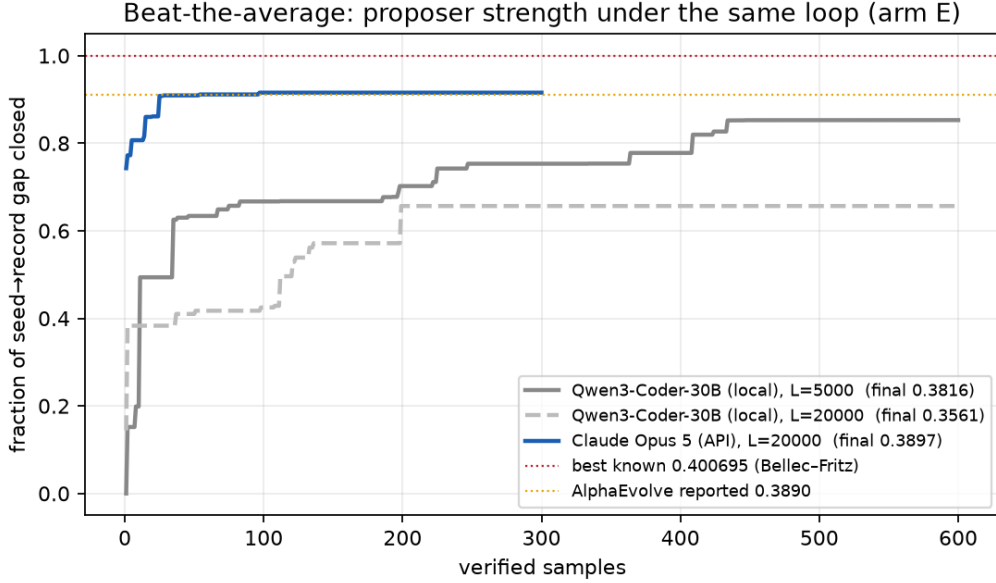


Figure 1: Beat-the-average under the same loop (arm E): fraction of the seed-to-record gap closed per verified sample. The local 30B at two grids; Claude Opus 5; dotted lines mark AlphaEvolve’s reported value and the best known (Bellec-Fritz).

and a characterized wall — grid, time budget and proposer strength all ruled out — leaving the construction family itself.

2 Related work

[TODO: Condense from the review: FunSearch/AlphaEvolve/ShinkaEvolve/EoH/ReEvo (scaffolding, not operator factorials); Mutation Without Variation (attractors without selection); Large Language Monkeys (coverage law); pass@k inversion literature; ReEvo reflections vs our persistent schematic notebook; the bin-packing audits.]

3 The harness and the problems

[TODO: Describe frontier_search.py: islands, elites, migration; chat-mode proposing; sandboxed verification with behaviour signatures; the operators (memory/agenda/novelty/repel) exactly as run. The six problems ported from the official notebooks, with best-known values and seeds. One table.]

4 The factorial (battery F)

[TODO: Arms A–E; budget 120 samples; metrics frac/auc/div; the table by problem and arm (both replicates); pre-registered tests F1/F2/F4 + F-ext (F1’, F6, F4’, F7); the granularity-floor discussion (exact sign-flip at n=6); the heterogeneity reading (D bimodal by validity-manifold width).]

5 Scale: the record attempts

[TODO: The pre-registered ladder: E vs D at 600 samples (scaling inversion: 0.3816 vs 0.3514 from a 120-sample tie); grid change (L=20000 hurts the local proposer at equal budget); the frontier proposer (0.3897 at 300 samples, past AlphaEvolve's reported 0.3890, plateau from gen 8); the 60-second verifier budget (+0.0002). The legible constructions: the 30B's bimodal 16-atom measure with commented strategy; Claude's exact sparse-atom objective with analytic gradient and multiplicative-weights refinement. The wall: every proposer writes the sparse-atom family; the record construction lies elsewhere.]

6 Discussion

[TODO: (a) What office scale buys and where it stops (the calibration map); (b) operators: composition as the only safe arm; the anchor law across five measurements; (c) the proposer lever vs the idea wall; relation to "mass, not frontier" and the bit distance; (d) design rules for verified search at small scale.]

7 Limitations

[TODO: n=6 problems and the exactness floor; one domain for the record saga; API proposer nondeterminism; verifier time budgets; best-known values move (live repository); no head-to-head with ShinkaEvolve (open question; their system is open-source).]

8 Conclusion

[TODO: One paragraph: operators help and compose; proposers move the sample-efficiency curve by orders of magnitude; and every lever we could pull stalls at the same wall, which is conceptual. "The last mile is an idea, not a budget."]

References

- [1] Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner. Mathematical exploration and discovery at scale, 2025.
- [2] Roberto I. Ono Filho. Interrupting the loop: Periodic subject changes raise judged surprise and connection in base language models, 2026.